AYUSH DWIVEDI

B.Tech Undergraduate | MERN Stack & Full-Stack Developer

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GitHub: github.com/ayushdwivedi-stack | LinkedIn: linkedin.com/in/ayush-dwivedi | LeetCode (300+ Problems)

PROFESSIONAL SUMMARY

Driven B.Tech undergraduate at Galgotias College with strong practical expertise in full-stack MERN development (MongoDB, Express.js, React.js, Node.js), RESTful API engineering, and modern web application architecture. Active open-source contributor with multiple reviewed and merged pull requests across established repositories. Demonstrated ability to build client web solutions from requirements to production deployment with rigorous testing, code reviews, and Git workflows. Based in Greater Noida / Noida and immediately available for on-site client delivery and product development.

TECHNICAL SKILLS

- **MERN & Frontend:** React.js, Node.js, Express.js, MongoDB (Mongoose), JavaScript (ES6+), TypeScript, Tailwind CSS, HTML5, CSS3
- **Backend & APIs:** RESTful API Design, JWT Authentication, Role-Based Access Control (RBAC), WebSockets, Middleware Development
- **Databases & Caching:** MongoDB, PostgreSQL, Redis (Caching & Session Store), SQL, Schema Design & Query Optimization
- **DevOps & Tooling:** Git, GitHub, GitHub Actions (CI/CD), Postman, Docker, Linux/Bash, npm, pnpm, Vitest
- **Core Computer Science:** Data Structures & Algorithms (300+ LeetCode problems in C++), OOP, System Design Fundamentals

FEATURED PROJECTS

Enterprise MERN Collaborative Web Platform | *React.js · Node.js · Express · MongoDB · Redis · WebSockets · Tailwind CSS*

- Architected a scalable full-stack web application with responsive React.js UI, modular Express.js services, and MongoDB database models.
- Implemented secure JWT authentication, password hashing with bcrypt, role-based authorization (RBAC), and centralized API error handling.
- Integrated Redis caching and MongoDB query indexing, achieving sub-60ms response latencies during concurrent API stress testing.
- Engineered real-time bidirectional messaging channels using WebSockets for live collaboration across distributed client workspaces.

Wafer Defect Quality & Analytics Dashboard | *React.js · Node.js · Express.js · PyTorch · REST APIs · MongoDB*

- Developed an end-to-end full-stack analytics portal connecting an interactive React.js interface with high-throughput Node.js/Express backend APIs.
- Built robust file upload endpoints, multi-part form validation, and asynchronous processing pipelines for industrial defect classification.
- Designed real-time data visualization charts and telemetry summaries for manufacturing QA tracking, persisting history in MongoDB.

Valkey / Redis Admin & Infrastructure Dashboard | *React.js · Node.js · Express.js · Redis · REST APIs · Tailwind CSS*

- Engineered a web management console providing real-time telemetry, key-space monitoring, and memory fragmentation tracking for Redis instances.
- Implemented high-performance Express.js endpoints for cluster node health diagnostics and automated latency alerting.

Open-Source Contributions to Production Codebases | *React · WebSerial · Nuxt / Vitest · Bun · Node.js · Git Workflows*

- Authored and submitted verified production patches to established open-source repositories: meshtastic/web (React/WebSerial disconnect handling), meshtastic/web-flasher, and cryptostrapon/npm-install-audit.
- Maintained 100% test pass rates across automated test suites (Vitest, Bun test), actively collaborating through maintainer code reviews.

EDUCATION

Galgotias College of Engineering & Technology, Greater Noida

Bachelor of Technology (B.Tech) | CGPA: 8.0 / 10.0 | 2024 – 2028 | Noida / Greater Noida

ACHIEVEMENTS & COMPETITIVE HIGHLIGHTS

- HackIndia Spark 4 Finalist: Placed among top engineering teams, recognized for rapid full-stack MVP architecture and live prototype deployment.
- Competitive Programming: Solved 300+ problems on LeetCode & Codeforces with strong foundation in Data Structures, Algorithms, and C++ optimization.
- Open-Source Contributor: Active contributor with verified pull requests accepted into established developer tooling and web repositories.